Adil NAS

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- **GitHub:** https://github.com/Adilnasceng
- **LinkedIn:** https://www.linkedin.com/in/adilnas/
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About Me

Computer Engineer (B.Sc., Honors) and M.Sc. student at Sivas Cumhuriyet University, specializing in robotics, autonomous systems, and software development. Led the university's Teknofest team to 2nd place in Turkey with a fully autonomous robot, handling ROS2 software, SLAM, navigation, and sensor integration. Experienced in AI/ML (Samsung Innovation Campus), embedded systems, and full-stack development. Published researcher in IoT-enhanced autonomous robotic architectures.

Publications

NAS, A.; Yüksek, A. G.; Mandacı, E. D.
IoT-Enhanced MIS Architectures for Autonomous Robotic Systems. DOI

Education

Cumhuriyet University — M.Sc. in Computer Engineering **Sivas,**
GPA: 3.75 / 4.00 September 2025 – Present
Thesis: *Autonomous Mobile Robot Navigation System Based on Audio, Visual, and Spatial Sensor Data Fusion*

Cumhuriyet University — B.Sc. in Computer Engineering **Sivas,**
GPA: 3.03 / 4.00 - Honors Graduate September 2021 – June 2025

Experience

AI Trainee — Samsung Innovation Campus **Remote**
July – November 2025

- Completed structured AI training covering Machine Learning and Deep Learning fundamentals.
- Applied Python for data analysis, model building, and evaluation tasks.
- Gained hands-on experience with LLM orchestration tools including Ollama, CrewAI, and LangChain.
- Developed and presented an end-to-end AI project as part of the program capstone.

Teknofest — *Robotics and AI Team Lead* **Sivas,**
September 2024 – June 2025

- Led the development of an autonomous robot that achieved 2nd place in the Teknofest Digital Technologies in Industry Competition (Advanced Category).
- Implemented all ROS2-based software, including sensor integration, SLAM (SLAM Toolbox), and autonomous navigation (Nav2).
- Developed the entire software pipeline end-to-end; also contributed to hardware and electronics.
- Gained extensive experience in robotics, sensor fusion, and real-world autonomous systems.

C-Prot — *Software Intern* **Mersin, On-site**
July – August 2024

- Developed file search and malware detection algorithms in C++.
- Created a project focused on certificate scanning in files.

- Built a full-stack application using Svelte and Rust within the Tauri environment.

Projects

Autonomous Line-Following Robot

- Developed a mobile robot using Raspberry Pi 4 to follow a black stripe via onboard camera.
- Controlled DC motors through GPIO using a custom motor driver circuit.
- Trained a machine learning model with Roboflow to detect line splits and determine direction.
- Gained hands-on experience with embedded systems, motor control, and motion algorithms.

AGV with Natural Navigation and Mapping

- Built an AGV (Autonomous Guided Vehicle) using Raspberry Pi 4 and ROS2 Humble on Ubuntu.
- Implemented features such as SLAM-based mapping, obstacle avoidance, and restricted/low-speed zone detection using LIDAR.
- Integrated ROS2 packages like SLAM Toolbox, Nav2, and ROS2 Control.
- Designed URDF robot models and provided navigation capabilities for factory logistics scenarios.
- Established communication between Raspberry Pi and Arduino for motor control, encoder feedback, and IMU data processing.
- Gained strong experience in robotics, sensor fusion, autonomous navigation, and the ROS2 ecosystem.

Technical Skills

Programming Languages:
C++, Python, JavaScript, C#, Java, HTML/CSS, SQL

Technologies & Tools:
React, Node.js, Flask, Git/GitHub, Tauri, Svelte, Linux, Arduino, Raspberry Pi, Jetson, ROS2, OpenCV, Docker, Ollama, CrewAI and LangChain.

Protocols & Interfaces:
I2C, SPI, UART/USART, CAN, PWM, GPIO

Languages

- **English:** Intermediate (B1–B2)
- **Turkish:** Native

Soft Skills

- Quick Learner & Problem Solver
- Team Collaboration
- Curiosity for New Technologies
- Strong Research Skills

Interests

Software Development, Embedded Systems, Web Technologies, Robotics, Open-Source Projects.

References

Will be shared upon request.